

Posets in Sage - An Introduction

Math 737 - Lab 1

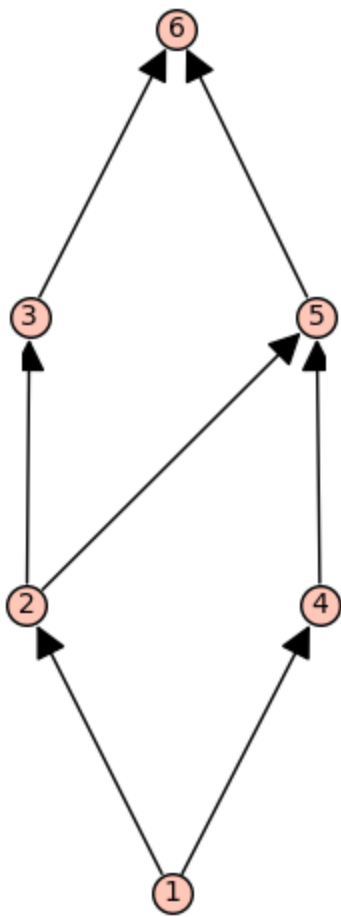
First define a poset.

Make a list of the elements of your poset, then a list of the covering relations between poset element.

Evaluate the cell below by putting your cursor in the block and then pressing shift + enter

```
In [3]: elem = [1,2,3,4,5,6]
rels = [[1,2],[2,3],[4,5],[5,6],[1,4],[2,5],[3,6]]
Q = Poset((elem,rels))
Q.plot()
```

Out[3]:

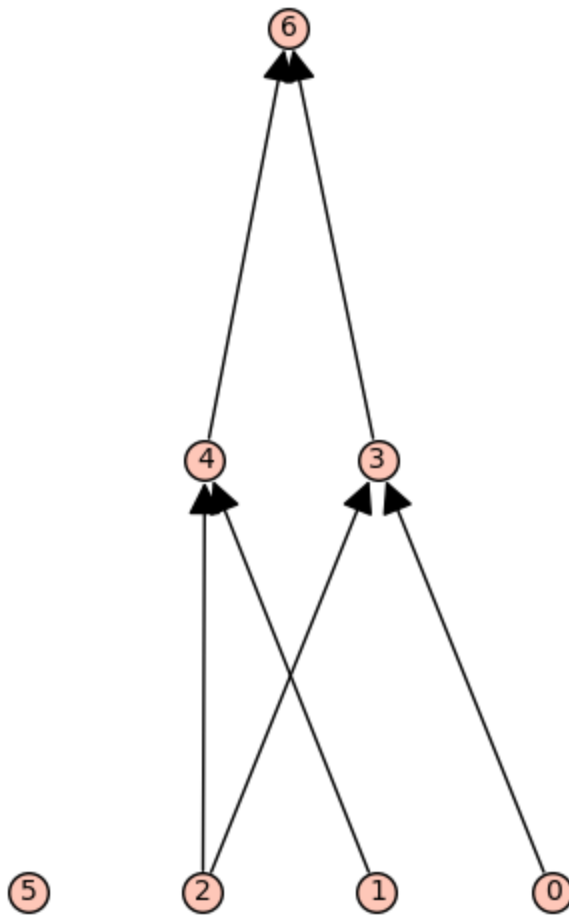


You can also make a random poset.

Evaluate the cell below multiple times to see multiple random posets of 7 elements.

```
In [4]: myposet = Posets(7).random_element()
        plot(myposet)
```

Out[4]:



Exercise 1: Make some of the example posets P, Q, R, S, T, N, from our last class. We already made Q above.

```
In [5]: # P
        elem = ['a', 'b', 'c', 'd', 'e']
        rels = [['a', 'b'], ['b', 'd'], ['d', 'e'], ['a', 'c'], ['c', 'e']]
        P = Poset((elem, rels))
        P.plot()

        # Q
        elem = ['a', 'b', 'c', 'd', 'e']
        rels = [['a', 'b'], ['a', 'c'], ['b', 'd'], ['c', 'd'], ['c', 'e'], ['d', 'f'], ['e', 'f']]
        Q = Poset((elem, rels))
        Q.plot()

        # R
        elem = ['a', 'b', 'c']
        rels = [['a', 'c'], ['b', 'c']]
        R = Poset((elem, rels))
        R.plot()
```

```

# S
elem = ['a', 'b', 'c', 'd']
rels = [['a', 'c'], ['a', 'd'], ['b', 'c'], ['b', 'd']]
S = Poset((elem,rels))
S.plot()

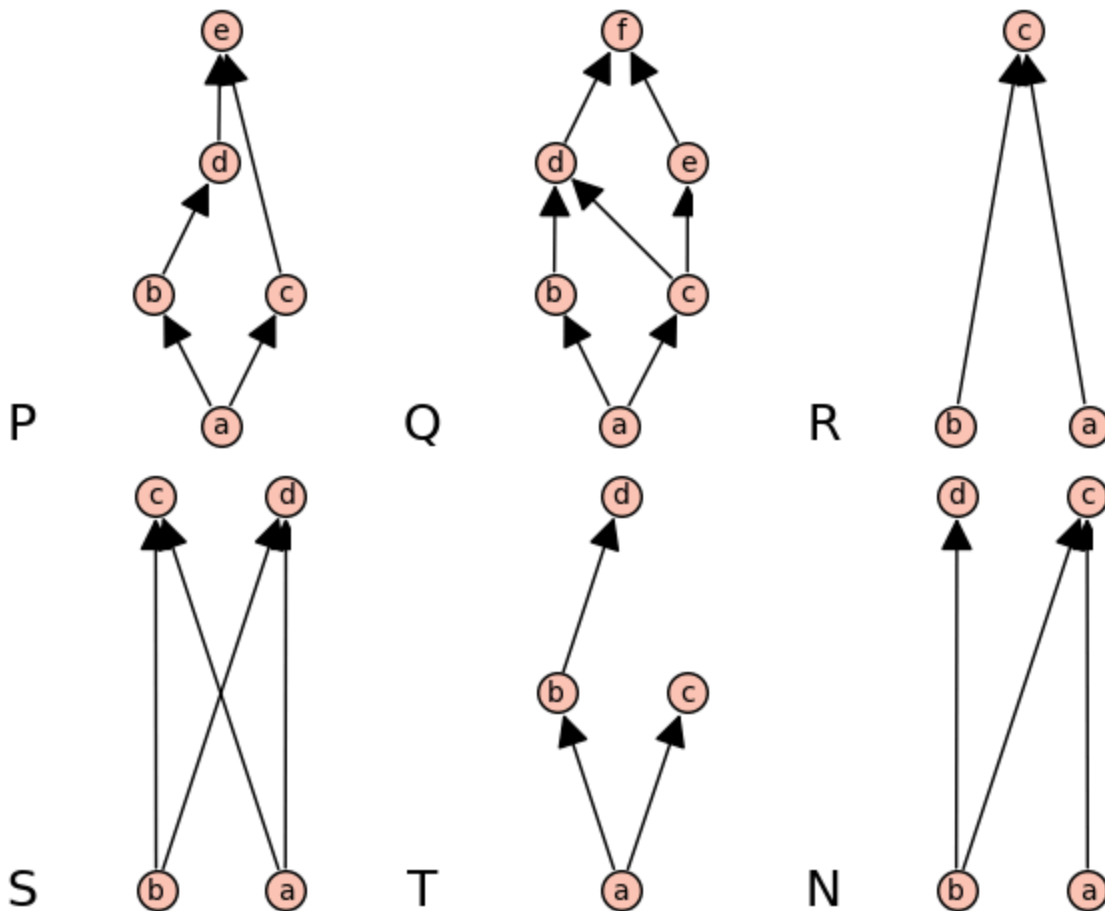
# T
elem = ['a', 'b', 'c', 'd']
rels = [['a', 'b'], ['b', 'd'], ['a', 'c']]
T = Poset((elem,rels))
T.plot()

# N
elem = ['a', 'b', 'c', 'd']
rels = [['a', 'c'], ['b', 'c'], ['b', 'd']]
N = Poset((elem,rels))
N.plot()

def L(X, name):
    g = X.plot(axes=False, figsize=4)
    return g + text(name, (0,0), fontsize=18, color='black')

graphics_array([
    [L(P, "P"), L(Q, "Q"), L(R, "R")],
    [L(S, "S"), L(T, "T"), L(N, "N")]
]).show()

```



Exercise 2: Use Sage to verify your answers from yesterday's class. You may need to access the documentation to see what the function is actually doing.

```
In [6]: # Disjoint sum (disjoint union) of T and R
PdQ = Q.disjoint_union(P)
# Ordinal sum of P and Q
PoQ = P.ordinal_sum(Q)
# Cartesian product of P and Q
PxQ = P.product(Q)
# Ordinal Product of P and Q
PoxQ = P.ordinal_product(Q)
```

```
In [7]: g1 = PdQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PdQ}
)

g2 = PoQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PoQ}
)

g3 = PxQ.plot(
```

```

    axes=False,
    element_labels={v: v[1] for v in PxQ}
)

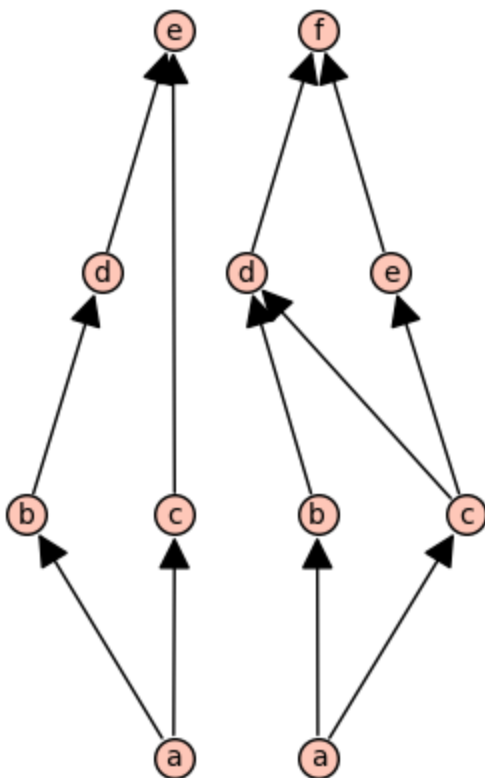
g4 = PoxQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PoxQ}
)

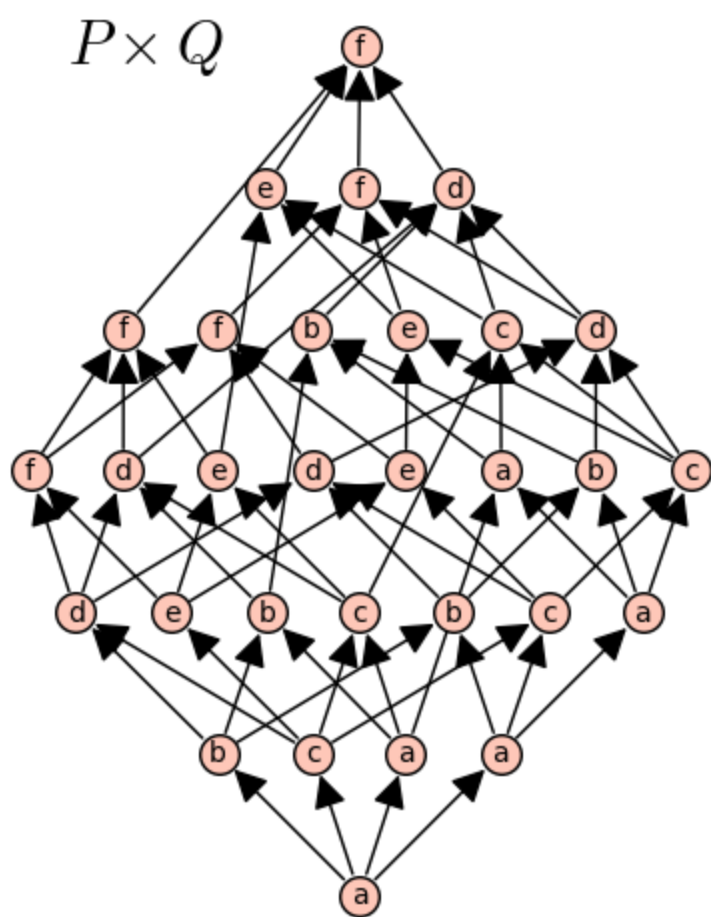
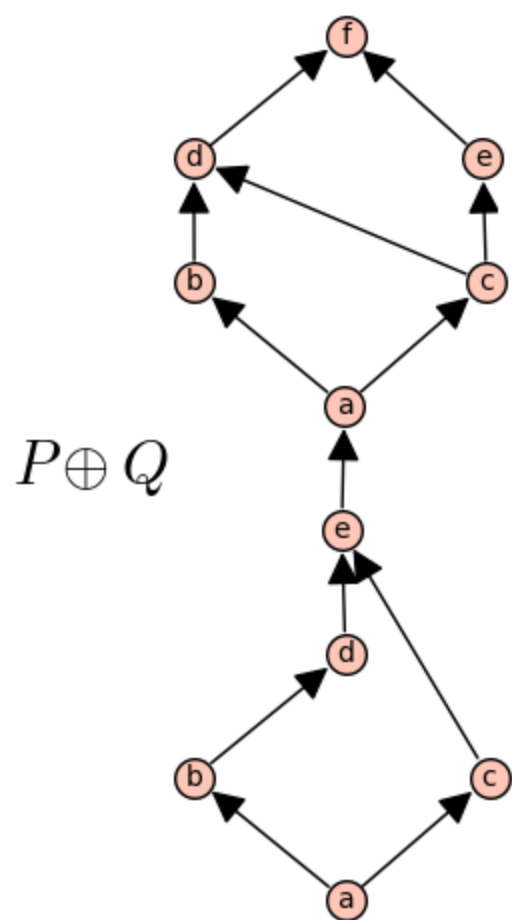
h = PoQ.height() # This is a constant to scale things
g1.set_axes_range(-h, h, -1, h + 0.5)

(g1 + text(r"$P + Q$", (1.5, 3.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g2 + text(r"$P \oplus Q$", (1.5, 3.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g3 + text(r"$P \times Q$", (1.5, 6),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g4 + text(r"$P \otimes Q$", (1.5, 10),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()

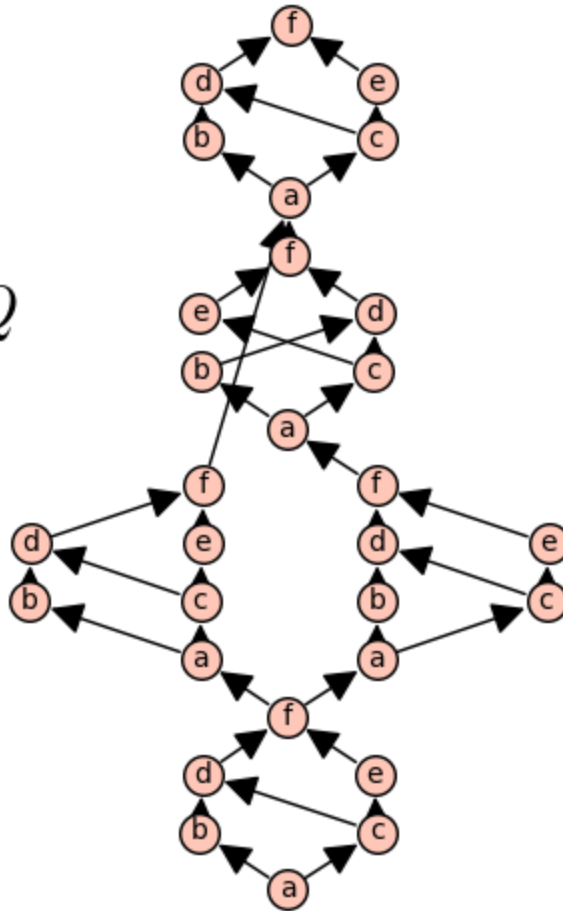
```

$P + Q$





$P \otimes Q$



The following function is code for the rank generating function of a poset P. It will not work if your poset does not have a rank function.

Evaluate the cell to define the function and then use it to find the rank generating functions of our example posets.

```
In [8]: def rank_generating_function(P):
        if not P.is_ranked():
            return "Does not exist, poset is not ranked"
        f = P.rank_function()
        rk = P.rank()
        list_of_ranks = [f(i) for i in P]
        R.<x> = PolynomialRing(QQ)
        rgf = 0
        for elem in list_of_ranks:
            rgf += x^elem
        return rgf

# Poset dictionary
posets = {
    "P": P,
    "Q": Q,
    "R": R,
    "S": S,
    "T": T,
```

```

    "N": N
}

for key,poset in posets.items():
    print(f"Rank generating function of {key}: {rank_generating_function(poset)}")

```

Rank generating function of P: Does not exist, poset is not ranked
 Rank generating function of Q: $x^3 + 2x^2 + 2x + 1$
 Rank generating function of R: $x + 2$
 Rank generating function of S: $2x + 2$
 Rank generating function of T: $x^2 + 2x + 1$
 Rank generating function of N: $2x + 2$

An order ideal is a subset of poset elements that is closed downward.

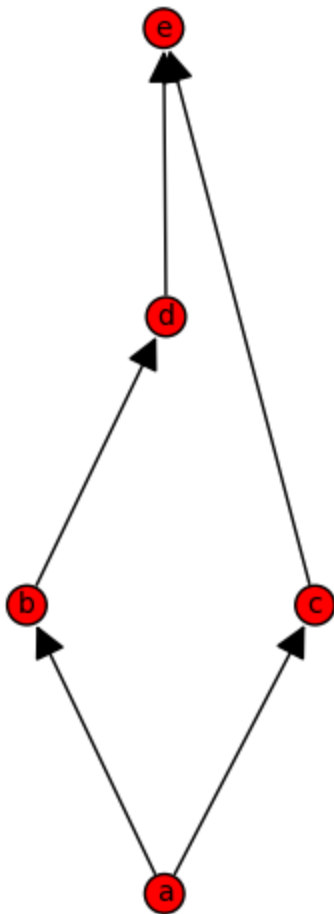
Let's make a random order ideal, using the code below. The green elements are the order ideal. The red elements are an order filter.

```

In [9]: oi = P.random_order_ideal()
        list(oi)
        P.order_ideal_plot(oi)

```

Out[9]:



Here is one of my favorite posets: the triangle

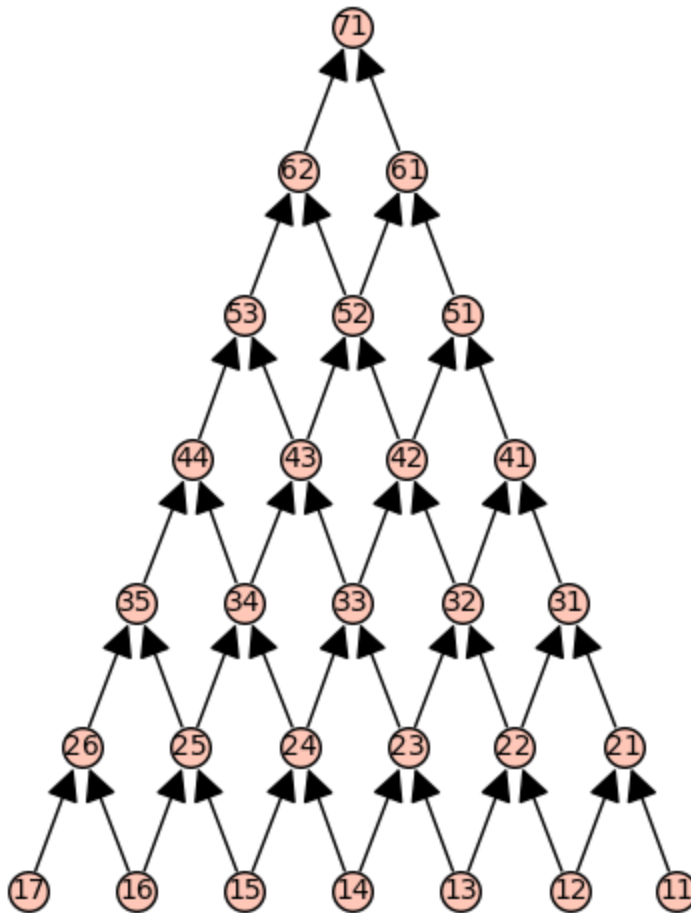

```
In [10]: elemP7 = [11, 12, 13, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34,
relP7 = [[11,21],[12, 21],[12, 22],[13, 22],[13, 23],[14, 23],[14, 24],[15,2
P7 = Poset((elemP7,relP7))
```

Evaluate the following cells to see some ways to plot posets

The normal way

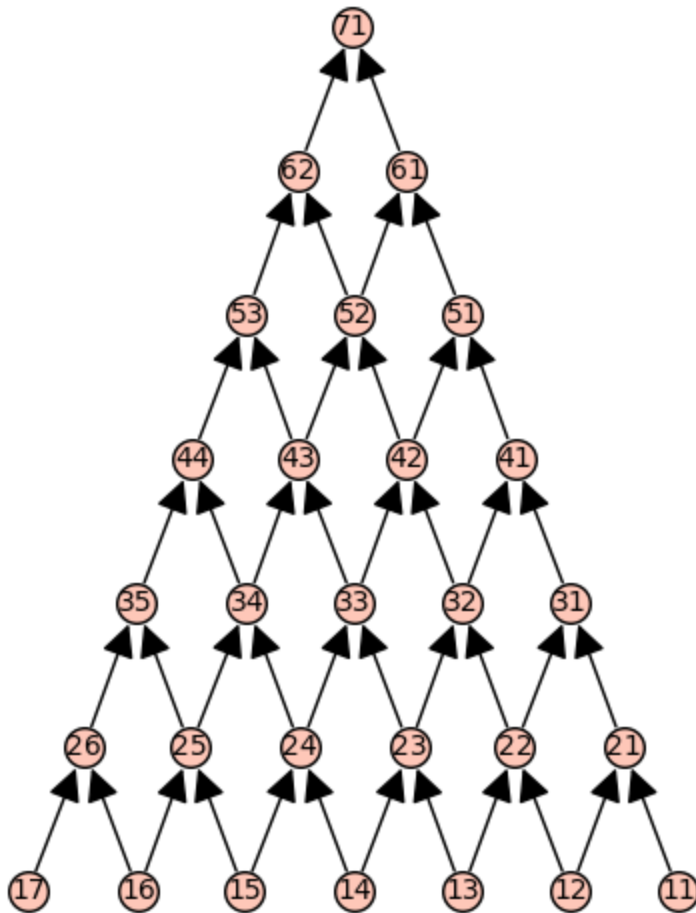
```
In [11]: plot(P7)
```

Out[11]:



The pretty way

```
In [12]: show(P7.plot())
```



Latex code for putting in your paper

In [13]: `latex(P7)`

Warning: `tikz.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit '...'.

Warning: `tkz-graph.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit 'https://www.ctan.org/pkg/tkz-graph'.

Warning: `tkz-berge.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit 'https://www.ctan.org/pkg/tkz-berge'.

```
Out[13]: \begin{tikzpicture}
\definecolor{cv0}{rgb}{0.0,0.0,0.0}
\definecolor{cfv0}{rgb}{1.0,1.0,1.0}
\definecolor{clv0}{rgb}{0.0,0.0,0.0}
\definecolor{cv1}{rgb}{0.0,0.0,0.0}
\definecolor{cfv1}{rgb}{1.0,1.0,1.0}
\definecolor{clv1}{rgb}{0.0,0.0,0.0}
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\definecolor{cv18}{rgb}{0.0,0.0,0.0}
```

```
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```

```

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%
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\Vertex[style={minimum size=1.0cm,draw=cv14,fill=cfv14,text=clv14,shape=circle},LabelOut=false,L=\hbox{$53$},x=0.105cm,y=1.611cm]{v14}
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\Vertex[style={minimum size=1.0cm,draw=cv18,fill=cfv18,text=clv18,shape=circle},LabelOut=false,L=\hbox{$42$},x=1.8083cm,y=0.8651cm]{v18}
\Vertex[style={minimum size=1.0cm,draw=cv19,fill=cfv19,text=clv19,shape=circle},LabelOut=false,L=\hbox{$52$},x=0.9971cm,y=0.7162cm]{v19}

```

```

\Vertex[style={minimum size=1.0cm,draw=cv20,fill=cfv20,text=clv20,shape=cir
cle},LabelOut=false,L=\hbox{$62$},x=0.2664cm,y=0.7578cm]{v20}
\Vertex[style={minimum size=1.0cm,draw=cv21,fill=cfv21,text=clv21,shape=cir
cle},LabelOut=false,L=\hbox{$11$},x=5.0cm,y=0.537cm]{v21}
\Vertex[style={minimum size=1.0cm,draw=cv22,fill=cfv22,text=clv22,shape=cir
cle},LabelOut=false,L=\hbox{$21$},x=4.3803cm,y=0.6666cm]{v22}
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\Vertex[style={minimum size=1.0cm,draw=cv24,fill=cfv24,text=clv24,shape=cir
cle},LabelOut=false,L=\hbox{$41$},x=2.7955cm,y=0.3894cm]{v24}
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cle},LabelOut=false,L=\hbox{$51$},x=1.9527cm,y=0.1224cm]{v25}
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cle},LabelOut=false,L=\hbox{$71$},x=0.4806cm,y=0.0012cm]{v27}
%
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\Edge[lw=0.1cm,style={post, bend right,color=cv1v2,,},](v1)(v2)
\Edge[lw=0.1cm,style={post, bend right,color=cv1v4,,},](v1)(v4)
\Edge[lw=0.1cm,style={post, bend right,color=cv2v5,,},](v2)(v5)
\Edge[lw=0.1cm,style={post, bend right,color=cv3v4,,},](v3)(v4)
\Edge[lw=0.1cm,style={post, bend right,color=cv3v7,,},](v3)(v7)
\Edge[lw=0.1cm,style={post, bend right,color=cv4v5,,},](v4)(v5)
\Edge[lw=0.1cm,style={post, bend right,color=cv4v8,,},](v4)(v8)
\Edge[lw=0.1cm,style={post, bend right,color=cv5v9,,},](v5)(v9)
\Edge[lw=0.1cm,style={post, bend right,color=cv6v7,,},](v6)(v7)
\Edge[lw=0.1cm,style={post, bend right,color=cv6v11,,},](v6)(v11)
\Edge[lw=0.1cm,style={post, bend right,color=cv7v8,,},](v7)(v8)
\Edge[lw=0.1cm,style={post, bend right,color=cv7v12,,},](v7)(v12)
\Edge[lw=0.1cm,style={post, bend right,color=cv8v9,,},](v8)(v9)
\Edge[lw=0.1cm,style={post, bend right,color=cv8v13,,},](v8)(v13)
\Edge[lw=0.1cm,style={post, bend right,color=cv9v14,,},](v9)(v14)
\Edge[lw=0.1cm,style={post, bend right,color=cv10v16,,},](v10)(v16)
\Edge[lw=0.1cm,style={post, bend right,color=cv10v11,,},](v10)(v11)
\Edge[lw=0.1cm,style={post, bend right,color=cv11v17,,},](v11)(v17)
\Edge[lw=0.1cm,style={post, bend right,color=cv11v12,,},](v11)(v12)
\Edge[lw=0.1cm,style={post, bend right,color=cv12v18,,},](v12)(v18)
\Edge[lw=0.1cm,style={post, bend right,color=cv12v13,,},](v12)(v13)
\Edge[lw=0.1cm,style={post, bend right,color=cv13v19,,},](v13)(v19)
\Edge[lw=0.1cm,style={post, bend right,color=cv13v14,,},](v13)(v14)
\Edge[lw=0.1cm,style={post, bend right,color=cv14v20,,},](v14)(v20)
\Edge[lw=0.1cm,style={post, bend right,color=cv15v16,,},](v15)(v16)
\Edge[lw=0.1cm,style={post, bend right,color=cv15v22,,},](v15)(v22)
\Edge[lw=0.1cm,style={post, bend right,color=cv16v17,,},](v16)(v17)
\Edge[lw=0.1cm,style={post, bend right,color=cv16v23,,},](v16)(v23)
\Edge[lw=0.1cm,style={post, bend right,color=cv17v18,,},](v17)(v18)
\Edge[lw=0.1cm,style={post, bend right,color=cv17v24,,},](v17)(v24)
\Edge[lw=0.1cm,style={post, bend right,color=cv18v19,,},](v18)(v19)
\Edge[lw=0.1cm,style={post, bend right,color=cv18v25,,},](v18)(v25)
\Edge[lw=0.1cm,style={post, bend right,color=cv19v20,,},](v19)(v20)
\Edge[lw=0.1cm,style={post, bend right,color=cv19v26,,},](v19)(v26)
\Edge[lw=0.1cm,style={post, bend right,color=cv20v27,,},](v20)(v27)
\Edge[lw=0.1cm,style={post, bend right,color=cv21v22,,},](v21)(v22)
\Edge[lw=0.1cm,style={post, bend right,color=cv22v23,,},](v22)(v23)
\Edge[lw=0.1cm,style={post, bend right,color=cv23v24,,},](v23)(v24)

```

```

\Edge[lw=0.1cm,style={post, bend right,color=cv24v25,,}(v24)(v25)
\Edge[lw=0.1cm,style={post, bend right,color=cv25v26,,}(v25)(v26)
\Edge[lw=0.1cm,style={post, bend right,color=cv26v27,,}(v26)(v27)
%
\end{tikzpicture}

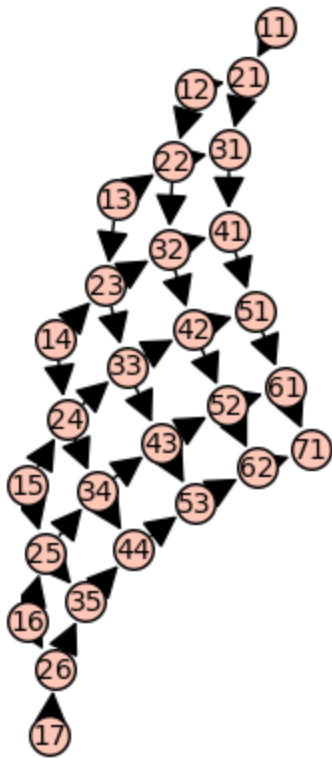
```

The boingy way

```

In [14]: H7 = P7.hasse_diagram()
show(H7)

```

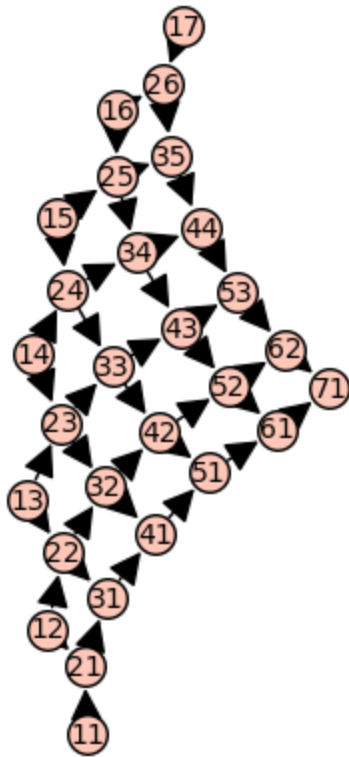


The sticky way

```

In [15]: H7.plot(save_pos=True)
show(H7)

```



Exercise 3: Use these various plot methods on our example posets. Which plot method is your favorite?

In [16]: I like boingy

Cell In[16], line 1

I like boingy

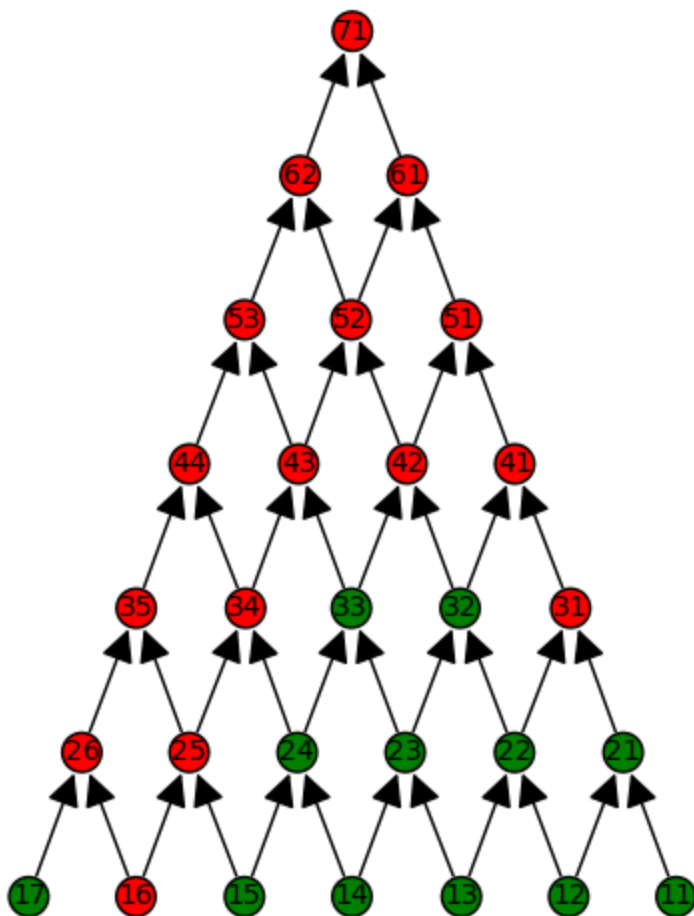
^

SyntaxError: invalid syntax

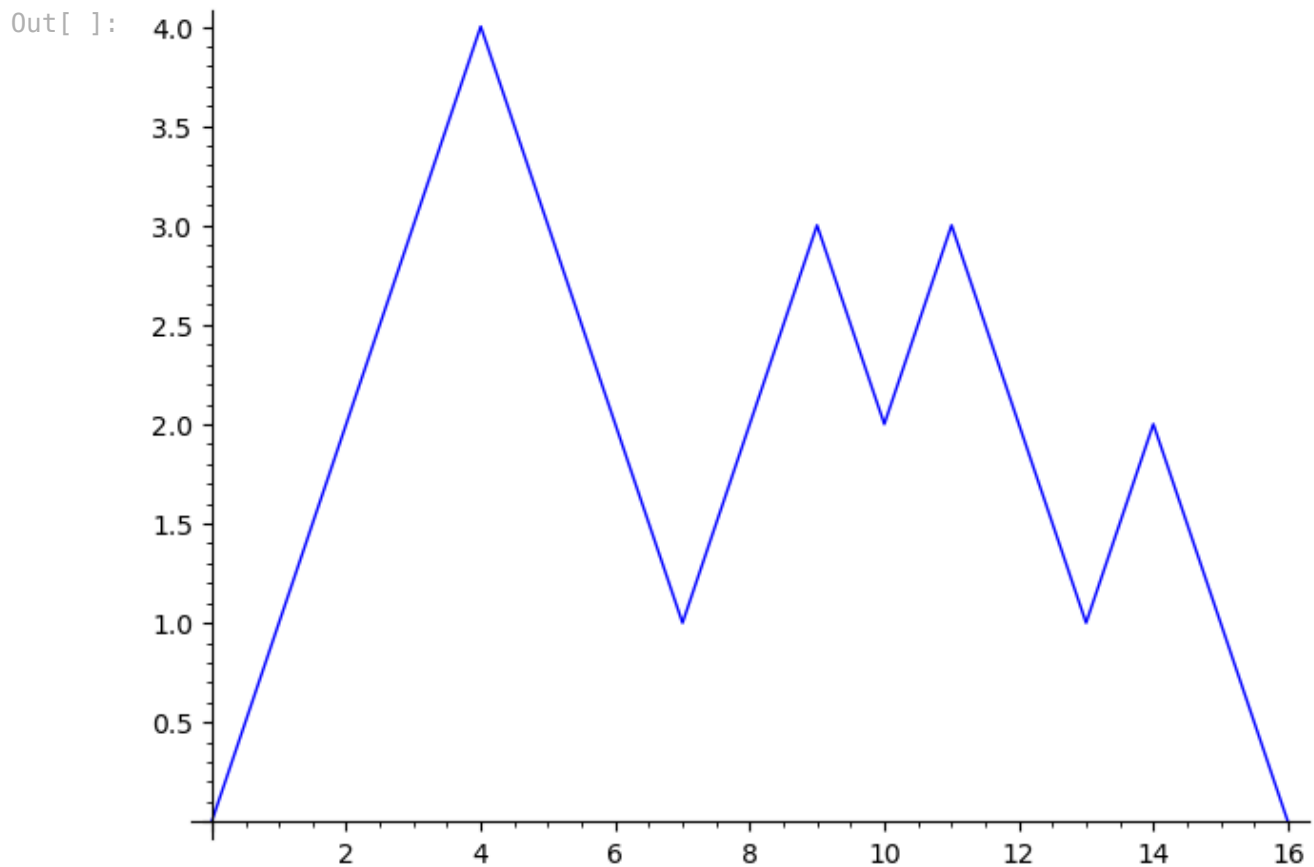
The reason the triangle poset is one of my favorite posets is that order ideals of the triangle poset are Dyck paths!

```
In [ ]: P7oi = P7.random_order_ideal()
        P7.order_ideal_plot(P7oi)
```


Out[]:



```
In [ ]: d = DyckWords(8).random_element()
plot(d)
```



We can use Sage to construct new posets from old

Exercise 4: Using our example posets P,Q,R,S,T,N, construct new posets as examples of the disjoint sum, ordinal sum, and direct (cartesian) product. Plot the new posets you create.

```
In [ ]: # Poset dictionary
P0 = {
    "P": P,
    "Q": Q,
    "R": R,
    "S": S,
    "T": T,
    "N": N
}

"""
examples of poset operations:

# Disjoint sum (disjoint union) of T and R
PdQ = Q.disjoint_union(P)
# Ordinal sum of P and Q
PoQ = P.ordinal_sum(Q)
```

```

# Cartesian product of P and Q
PxQ = P.product(Q)
# Ordinal Product of P and Q
PoxQ = P.ordinal_product(Q)
"""

RdT = R.disjoint_union(T)
RoN = R.ordinal_sum(N)
NxR = N.product(R)
NoxR = N.ordinal_product(R)

```

```

In [18]: g1 = RdT.plot(
    axes=False,
    element_labels={v: v[1] for v in RdT}
)

g2 = RoN.plot(
    axes=False,
    element_labels={v: v[1] for v in RoN}
)

g3 = NxR.plot(
    axes=False,
    element_labels={v: v[1] for v in NxR}
)

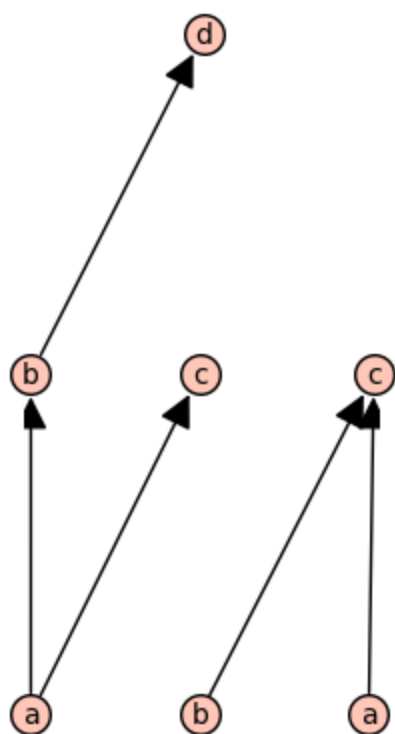
g4 = NoxR.plot(
    axes=False,
    element_labels={v: v[1] for v in NoxR}
)

h = PoQ.height() # This is a constant to scale things
g1.set_axes_range(-h, h, -1, h + 0.5)

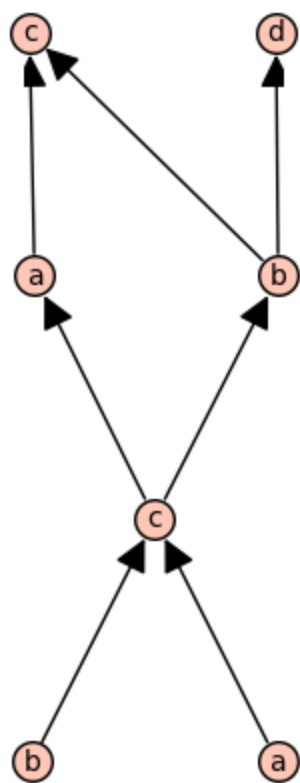
(g1 + text(r"$R + T$", (1, 2.5),
    fontsize=22, color='black',
    horizontal_alignment='center')).show()
(g2 + text(r"$R \oplus N$", (1.5, 3.5),
    fontsize=22, color='black',
    horizontal_alignment='center')).show()
(g3 + text(r"$N \times R$", (1, 2.5),
    fontsize=22, color='black',
    horizontal_alignment='center')).show()
(g4 + text(r"$N \otimes R$", (1.5, 3.5),
    fontsize=22, color='black',
    horizontal_alignment='center')).show()

```

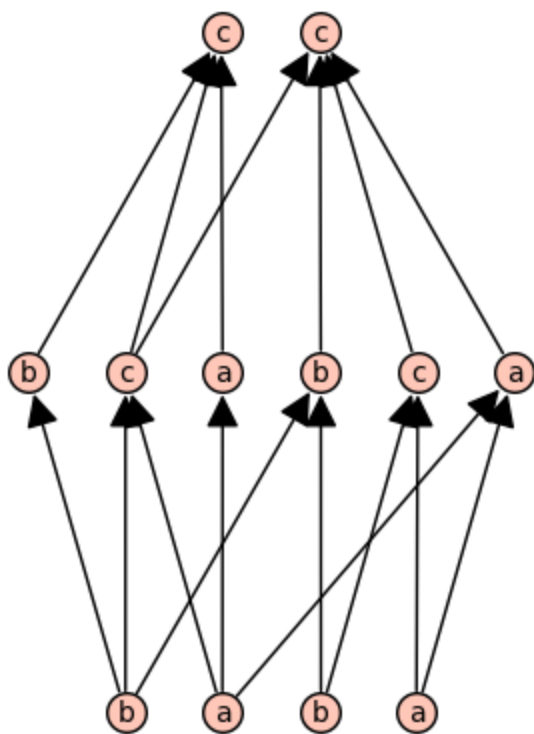
$$R + T$$



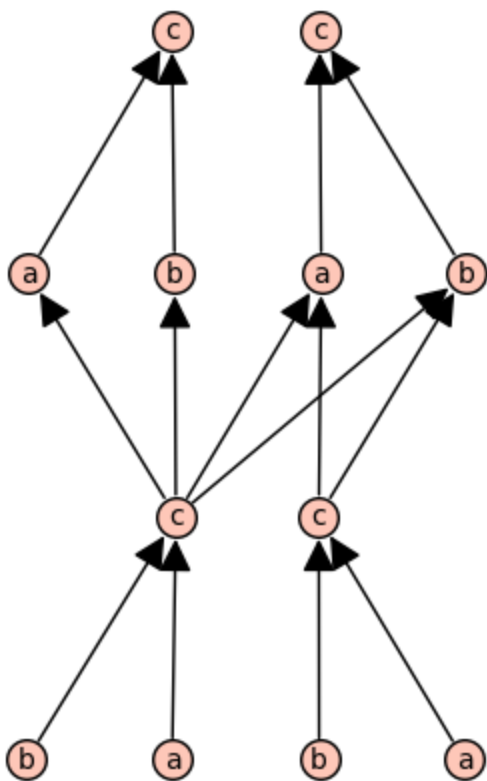
$$R \oplus N$$



$$N \times R$$



$$N \otimes R$$



In [0]: